

Name - Bhavya Goyal

Sap id - 500122949

Roll no - R2142231839

Batch - 1 (Full Stack AI)

School - School of Computer Science

Class Test - 2

- 1.) A 3-phase induction motor works on the principle of Electromagnetic Induction.
- When a 3-phase AC supply is given to the stator, it produces a rotating magnetic field (RMF).
 - This rotating field cuts the ~~rotor~~ rotor conductors.
 - Due to this, EMF is induced in the rotor (Faraday's Law).
 - Since the rotor circuit is closed, current flows in the rotor.
 - The interaction between rotor current and rotating magnetic field produces torque, causing the rotor to rotate.

It is called an asynchronous machine because rotor speed is ^{not} equal to synchronous speed. The rotor never reaches synchronous speed. If it did, there would be :-

- No relative motion.
- No induced EMF.
- No torque.

So, it always runs at slightly lower speed, hence called asynchronous.